

Лабораторная работа № 10. Настройка списков управления доступом (ACL)

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Цель работы

Освоить настройку прав доступа пользователей к ресурсам сети.

Задание

Задание

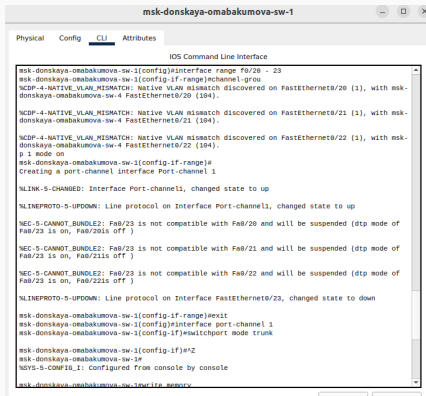
1. Требуется настроить следующие правила доступа:
 - web-сервер: разрешить доступ всем пользователям по протоколу HTTP через порт 80 протокола TCP, а для администратора открыть доступ по протоколам Telnet и FTP;
 - файловый сервер: с внутренних адресов сети доступ открыт по портам для общедоступных каталогов, с внешних — доступ по протоколу FTP;
 - почтовый сервер: разрешить пользователям работать по протоколам SMTP и POP3 (соответственно через порты 25 и 110 протокола TCP), а для администратора — открыть доступ по протоколам Telnet и FTP;
 - DNS-сервер: открыть порт 53 протокола UDP для доступа из внутренней сети;
 - разрешить icmp-сообщения, направленные в сеть серверов;

Задание

- разрешить доступ в сеть управления сетевым оборудованием только администратору сети.
 - запретить для сети Other любые запросы за пределы сети, за исключением администратора;
2. Требуется проверить правильность действия установленных правил доступа.
 3. Требуется выполнить задание для самостоятельной работы по настройке прав доступа администратора сети на Павловской.
 4. При выполнении работы необходимо учитывать соглашение об именовании.

Выполнение лабораторной работы

Ошибки предыдущей лабораторной работы



The screenshot shows a Cisco IOS Command Line Interface window titled "msk-donskaya-omabakumova-sw-1". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The CLI shows the following commands and their outputs:

```
msk-donskaya-omabakumova-sw-1(config)#interface range fa0/20 - 23
msk-donskaya-omabakumova-sw-1(config-if-range)#channel-group
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/20 (1), with msk-donskaya-omabakumova-sw-4 FastEthernet0/20 (104).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/21 (1), with msk-donskaya-omabakumova-sw-4 FastEthernet0/21 (104).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/22 (1), with msk-donskaya-omabakumova-sw-4 FastEthernet0/22 (104).
p 1 mode on
msk-donskaya-omabakumova-sw-1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINK-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/20 and will be suspended (dtp mode of Fa0/23 is on, Fa0/20 is off)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/21 and will be suspended (dtp mode of Fa0/23 is on, Fa0/21 is off)

%EC-5-CANNOT_BUNDLE2: Fa0/23 is not compatible with Fa0/22 and will be suspended (dtp mode of Fa0/23 is on, Fa0/22 is off)

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

msk-donskaya-omabakumova-sw-1(config-if-range)#exit
msk-donskaya-omabakumova-sw-1(config)#interface port-channel 1
msk-donskaya-omabakumova-sw-1(config-if)#switchport mode trunk

msk-donskaya-omabakumova-sw-1(config-if)#^Z
msk-donskaya-omabakumova-sw-1#
%SYS-5-CONFIG_I: Configured from console by console
msk-donskaya-omabakumova-sw-1#write memory
```

Рис. 1: Агрегирование

Ошибки предыдущей лабораторной работы

```
hostname msk-donskaya-omabakumova-sw-1
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXXK7m0
!
!
ip domain-name donsкаya.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXXK7m0
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
spanning-tree vlan 3 priority 24576
!
interface Port-channel1
 switchport mode trunk
!
interface FastEthernet0/1
!
interface FastEthernet0/2
!
interface FastEthernet0/3
!
interface FastEthernet0/4
```

Рис. 2: Добавлен интерфейс Port-channel1

Ошибки предыдущей лабораторной работы

```
.
interface FastEthernet0/20
switchport mode trunk
switchport nonegotiate
channel-group 1 mode on
!
interface FastEthernet0/21
switchport mode trunk
switchport nonegotiate
channel-group 1 mode on
!
interface FastEthernet0/22
switchport mode trunk
switchport nonegotiate
channel-group 1 mode on
!
interface FastEthernet0/23
switchport mode trunk
switchport nonegotiate
channel-group 1 mode on
!
interface FastEthernet0/24
switchport mode trunk
!
interface GigabitEthernet0/1
switchport mode trunk
,
```

Рис. 3: При отключении DTP появляется строчка свидетельствующая это

Ошибки предыдущей лабораторной работы

```
interface FastEthernet0/20
switchport mode trunk
channel-group 1 mode on
!
interface FastEthernet0/21
switchport mode trunk
channel-group 1 mode on
!
interface FastEthernet0/22
switchport mode trunk
channel-group 1 mode on
!
interface FastEthernet0/23
switchport mode trunk
channel-group 1 mode on
!
interface FastEthernet0/24
switchport mode trunk
!
interface GigabitEthernet0/1
switchport mode trunk
!
interface GigabitEthernet0/2
switchport mode trunk
```

Рис. 4: DTP включен, агрегирование завершено

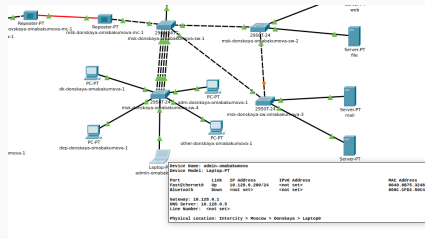


Рис. 5: Размещение ноутбука администратора в сети

Настройка правил доступа

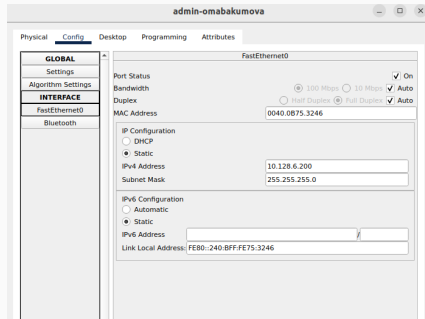


Рис. 6: Указание статического адреса

Настройка правил доступа

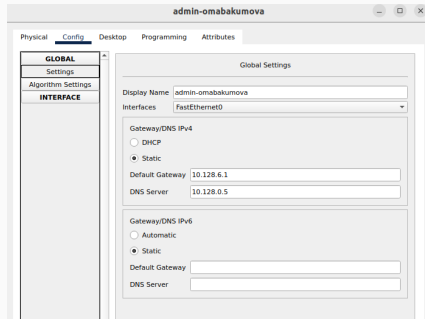


Рис. 7: Задание gateway-адреса и адреса DNS-сервера

Настройка правил доступа

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: {default port}

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::240:8FF:FE75:3246
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 10.128.0.200
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                10.128.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ping 10.128.0.5

Pinging 10.128.0.5 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.0.5

Pinging 10.128.0.5 with 32 bytes of data:

Reply from 10.128.0.5: bytes=32 time=1ms TTL=127
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127
Reply from 10.128.0.5: bytes=32 time=1ms TTL=127

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Рис. 8: Проверка конфигурации оконечного устройства и пингование

Настройка правил доступа



```
msk-donskaya-omabakumova-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

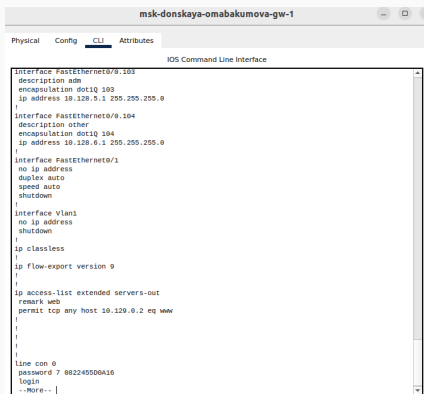
Press RETURN to get started:

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.101, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.102, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.103, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.104, changed state to up

User Access Verification
Password:
msk-donskaya-omabakumova-gw-1>en
Password:
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#ip access-list extended servers-out
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark web
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.2 eq 80
% Invalid input detected at '^' marker.
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp any host 10.129.0.2 eq 80
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#
```

Рис. 9: Настройка доступа к web-серверу по порту tcp 80

Настройка правил доступа



The screenshot shows a Cisco IOS Command Line Interface window titled "msk-donskaya-omabakumova-gw-1". The "CLI" tab is selected. The configuration displayed is as follows:

```
Interface FastEthernet0/0.103
description adm
encapsulation dot1Q 103
ip address 10.128.5.1 255.255.255.0
!
Interface FastEthernet0/0.104
description other
encapsulation dot1Q 104
ip address 10.128.6.1 255.255.255.0
!
Interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
Interface Vlan1
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended servers-out
remark web
permit tcp any host 10.129.0.2 eq www
!
!
!
!
line con 0
password 7 08224550A16
login
--More--
```

Рис. 10: Проверка изменений после проведения настройки с помощью команды “sh ru”

```
msk-donskaya-omabakumova-gw-1# configure terminal
msk-donskaya-gw-1(config)# interface f0 /0.3
msk-donskaya-gw-1(config-subif)# ip access-group servers-out out
```

```
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.0.1: Destination host unreachable.
Reply from 10.128.0.1: Destination host unreachable.
Reply from 10.128.0.1: Destination host unreachable.
Reply from 10.128.0.1: Destination host unreachable.

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.128.0.5

Pinging 10.128.0.5 with 32 bytes of data:

Reply from 10.128.0.1: Destination host unreachable.
Reply from 10.128.0.1: Destination host unreachable.
Reply from 10.128.0.1: Destination host unreachable.
Reply from 10.128.0.1: Destination host unreachable.

Ping statistics for 10.128.0.5:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Рис. 11: Пинг не проходит

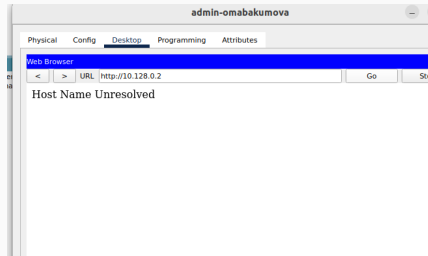


Рис. 12: Браузер тоже умер

Настройка правил доступа

```
USER ACCESS VET AT 10.128.0.101
Password:
msk-donskaya-omabakumova-gw-1>en
Password:
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#ip access-list extended servers-out
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp host 10.128.0.200 host 10.128.0.2
range 20 ftp
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp host 10.128.0.200 host 10.128.0.2 eq
telnet
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#^Z
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#
```

Copy Paste

Top

Рис. 13: Настройка дополнительного доступа администратора по протоколам Telnet и FTP

Настройка правил доступа

```
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended servers-out
remark web
permit tcp any host 10.129.0.2 eq www
permit tcp host 10.128.0.200 host 10.128.0.2 range 20 ftp
permit tcp host 10.128.0.200 host 10.128.0.2 eq telnet
!
!
!
!
line con 0
--More--
```

Copy Paste

Top

Рис. 14: Проверка результатов проведенной настройки по протоколам Telnet и FTP

Настройка правил доступа

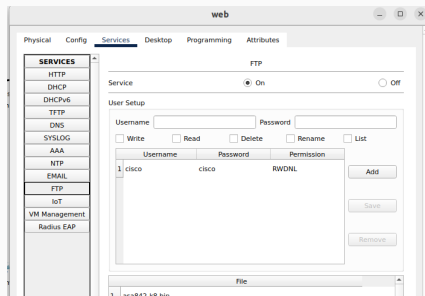


Рис. 15: FTP работает

Настройка правил доступа

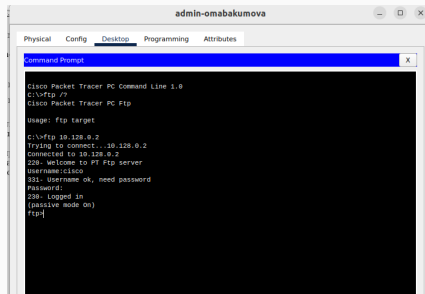


Рис. 16: Доступ есть по протоколу FTP

Настройка правил доступа

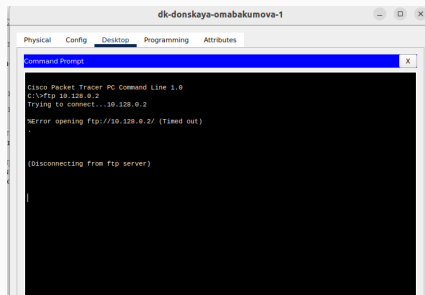


Рис. 17: Доступ запрещен при попытке с dk-donskaya-omabaumova

Настройка правил доступа

```
mik-donskaya-omabakumova-gw-1(config)#ip access-list extended servers-out
mik-donskaya-omabakumova-gw-1(config-ext-nacl)#remark file
mik-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp 10.128.0.0 0.0.255.255 host
10.128.0.3 eq 445
mik-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.3 range 20 ftp
mik-donskaya-omabakumova-gw-1(config-ext-nacl)#^Z
mik-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

mik-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
```

Рис. 18: Настройка доступа к файловому серверу

```
ip access-list extended servers-out
!
!
ip access-list extended servers-out
remark web
permit tcp any host 10.128.0.2 eq www
permit tcp host 10.128.0.200 host 10.128.0.2 range 20 ftp
permit tcp host 10.128.0.200 host 10.128.0.2 eq telnet
remark file
permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445
permit tcp any host 10.128.0.3 range 20 ftp
!
!
```

Рис. 19: Проверка результатов проведенной настройки доступа к файловому серверу

Настройка правил доступа

```
Enter configuration commands, one per line. End with CTRL/Z.
msk-donskaya-omabakumova-gw-1(config)#ip access-list extended servers-out
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#remark mail
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.4 eq smtp
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.4 eq pop3
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#^Z
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#sh ru
..
..
```

Рис. 20: Настройка доступа к почтовому серверу

Настройка правил доступа

```
msk-donskaya-onabakunova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakunova-gw-1(config)#ip access-list extended servers-out
msk-donskaya-onabakunova-gw-1(config-ext-nacl)#remark dns
msk-donskaya-onabakunova-gw-1(config-ext-nacl)#permit udp 10.128.0.0 0.0.255.255 host
10.128.0.5 eq 53
msk-donskaya-onabakunova-gw-1(config-ext-nacl)#^Z
msk-donskaya-onabakunova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakunova-gw-1#wr mem
Building configuration...
[OK]
```

Рис. 21: Настройка доступа к DNS-серверу

```
ip access-list extended servers-out
remark web
permit tcp any host 10.129.0.2 eq www
permit tcp host 10.128.0.200 host 10.128.0.2 range 20 ftp
permit tcp host 10.128.0.200 host 10.128.0.2 eq telnet
remark file
permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445
permit tcp any host 10.128.0.3 range 20 ftp
remark mail
permit tcp any host 10.128.0.4 eq smtp
permit tcp any host 10.128.0.4 eq pop3
remark dns
permit udp 10.128.0.0 0.0.255.255 host 10.128.0.5 eq domain
!
```

Рис. 22: Проверка результатов проведенной настройки для почтового и DNS серверов

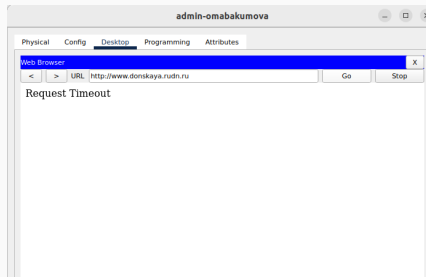


Рис. 23: Не работает

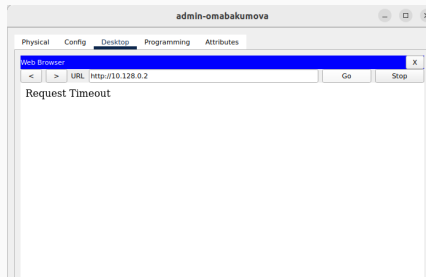


Рис. 24: Не работает 2.0

Настройка правил доступа

```
msk-donskaya-omabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-omabakumova-gw-1(config)#ip access-list extended servers-out
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#1 permit icmp any any
msk-donskaya-omabakumova-gw-1(config-ext-nacl)#2
msk-donskaya-omabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-omabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-omabakumova-gw-1#sh access-lists
Extended IP access list servers-out
 1 permit icmp any any
10 permit tcp any host 10.128.0.2 eq www
20 permit tcp host 10.128.0.290 host 10.128.0.2 range 20 ftp (8 match(es))
30 permit tcp host 10.128.0.290 host 10.128.0.2 eq telnet
40 permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445
50 permit tcp any host 10.128.0.3 range 20 ftp
60 permit tcp any host 10.128.0.4 eq smtp
70 permit tcp any host 10.128.0.4 eq pop3
80 permit udp 10.128.0.0 0.0.255.255 host 10.128.0.5 eq domain (2 match(es))

msk-donskaya-omabakumova-gw-1#
```

Рис. 25: Разрешение icmp-запросов

Настройка правил доступа

```
!
!
ip flow-export version 9
!
!
ip access-list extended servers-out
remark web
permit icmp any any
permit tcp any host 10.128.0.2 eq www
permit tcp host 10.128.0.200 host 10.128.0.2 range 20 ftp
permit tcp host 10.128.0.200 host 10.128.0.2 eq telnet
remark file
permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445
permit tcp any host 10.128.0.3 range 20 ftp
remark mail
permit tcp any host 10.128.0.4 eq smtp
permit tcp any host 10.128.0.4 eq pop3
remark dns
permit udp 10.128.0.0 0.0.255.255 host 10.128.0.5 eq domain
!
!
!
!
--More--
```

Рис. 26: Проверка результатов разрешения icstr-запросов

Настройка правил доступа

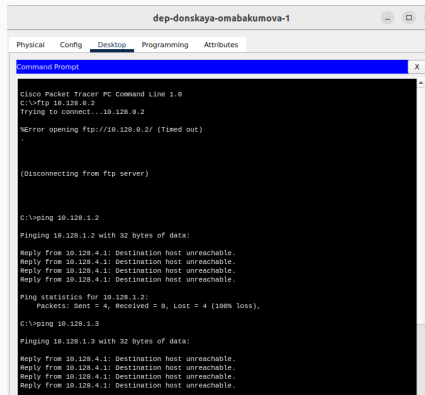
```
msk-donskaya-onabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-gw-1(config)#ip access-list extended other-in
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#remark admin
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit ip host 10.128.0.200 any
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#exit
msk-donskaya-onabakumova-gw-1(config)#interface fa/0/104
msk-donskaya-onabakumova-gw-1(config-subif)#ip access-group other-in in
msk-donskaya-onabakumova-gw-1(config-subif)#^Z
msk-donskaya-onabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-onabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-gw-1(config)#ip access-list extended management-out
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#remark admin
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit ip host 10.128.0.200 10.128.1.0 0.0.0.255
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#exit
msk-donskaya-onabakumova-gw-1(config)#interface fa/0/2
msk-donskaya-onabakumova-gw-1(config-subif)#ip access-group management-out out
msk-donskaya-onabakumova-gw-1(config-subif)#^Z
msk-donskaya-onabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-gw-1#wr mem
Building configuration...
[OK]
```

Рис. 27: Настройка доступа для сети Other и доступа администратора к сети сетевого оборудования

Самостоятельная работа



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'dep-donskaya-omabakumova-1'. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt. The Command Prompt shows the following sequence of commands and outputs:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2

%Error opening ftp://10.128.0.2/ (Timed out)

(Disconnecting from ftp server)

C:\>ping 10.128.1.2

Pinging 10.128.1.2 with 32 bytes of data:

Reply from 10.128.4.1: Destination host unreachable.
Reply from 10.128.4.1: Destination host unreachable.
Reply from 10.128.4.1: Destination host unreachable.
Reply from 10.128.4.1: Destination host unreachable.

Ping statistics for 10.128.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

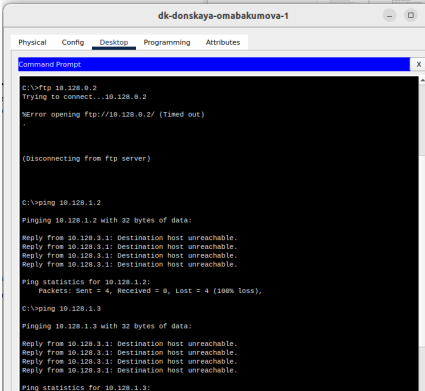
C:\>ping 10.128.1.3

Pinging 10.128.1.3 with 32 bytes of data:

Reply from 10.128.4.1: Destination host unreachable.
Reply from 10.128.4.1: Destination host unreachable.
Reply from 10.128.4.1: Destination host unreachable.
Reply from 10.128.4.1: Destination host unreachable.
```

Рис. 28: Проверка доступности с dep

Самостоятельная работа



```
dk-donskaya-omabakumova-1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
%Error opening ftp://10.128.0.2/ (Timed out)

(Disconnecting from ftp server)

C:\>ping 10.128.1.2
Pinging 10.128.1.2 with 32 bytes of data:

Reply from 10.128.3.1: Destination host unreachable.
Reply from 10.128.3.1: Destination host unreachable.
Reply from 10.128.3.1: Destination host unreachable.
Reply from 10.128.3.1: Destination host unreachable.

Ping statistics for 10.128.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.128.1.3
Pinging 10.128.1.3 with 32 bytes of data:

Reply from 10.128.3.1: Destination host unreachable.
Reply from 10.128.3.1: Destination host unreachable.
Reply from 10.128.3.1: Destination host unreachable.
Reply from 10.128.3.1: Destination host unreachable.

Ping statistics for 10.128.1.3:
```

Рис. 29: Проверка доступности с dk

Самостоятельная работа

```
admin-omabakumova
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
Connected to 10.128.0.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode on)
ftp>
C:\>ping 10.128.1.2
Pinging 10.128.1.2 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.128.1.2: bytes=32 time<1ms TTL=254
Reply from 10.128.1.2: bytes=32 time<1ms TTL=254
Ping statistics for 10.128.1.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.128.1.2
Pinging 10.128.1.2 with 32 bytes of data:
Reply from 10.128.1.2: bytes=32 time=3ms TTL=254
Reply from 10.128.1.2: bytes=32 time=24ms TTL=254
Reply from 10.128.1.2: bytes=32 time<1ms TTL=254
Reply from 10.128.1.2: bytes=32 time<1ms TTL=254
Ping statistics for 10.128.1.2:
```

Рис. 30: Проверка доступности с admin

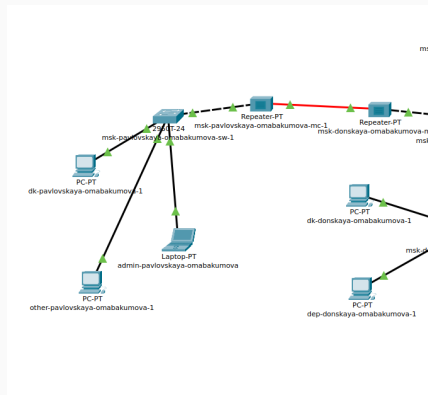


Рис. 31: Размещение нового устройства



Рис. 32: Устройство находится на Павловской

Самостоятельная работа

```
msk-donskaya-onabakumova-gw-1>en
Password:
msk-donskaya-onabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-gw-1(config)#ip access-list extended servers-out
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit tcp host 10.128.0.201 host 10.128.0.2 range 20 ftp
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit tcp host 10.128.0.201 host 10.128.0.2 range 20 ftp
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit tcp host 10.128.0.201 host 10.128.0.2 eq telnet
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#^Z
msk-donskaya-onabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-gw-1#wr m
Building configuration...
[OK]
msk-donskaya-onabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-gw-1(config)#ip access-list extended other-in
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#remark admin
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit ip host 10.128.0.201 any
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#exit
msk-donskaya-onabakumova-gw-1(config)#interface fa0/104
msk-donskaya-onabakumova-gw-1(config-subif)#ip access-group other-in in
msk-donskaya-onabakumova-gw-1(config-subif)#^Z
msk-donskaya-onabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-onabakumova-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-onabakumova-gw-1(config)#ip access-list extended management-out
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#remark admin
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit ip host 10.128.0.201
% Incomplete command.
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#permit ip host 10.128.0.201 10.128.1.0 0.0.0.255
msk-donskaya-onabakumova-gw-1(config-ext-nacl)#exit
msk-donskaya-onabakumova-gw-1(config)#interface fa0/2
msk-donskaya-onabakumova-gw-1(config-subif)#ip access-group management-out out
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-onabakumova-gw-1(config-subif)#ip access-group management-out out
msk-donskaya-onabakumova-gw-1(config-subif)#^Z
msk-donskaya-onabakumova-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-onabakumova-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-onabakumova-gw-1#
```

Рис. 33: Разрешение действий администратору из сети Other на Павловской

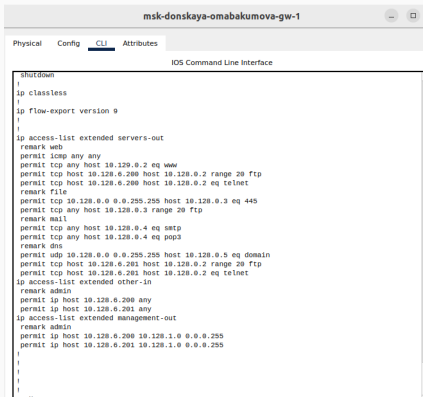


Рис. 34: Проверка результатов разрешения действий администратору из сети Other на Павловской

```
admin-pavlovskaya-omabukunova
Physical  Config  Sniffing  Programming  Attributes

Autonomous Mode!

Request timed out.
Reply from 10.120.1.2: bytes=32 time=0ms TTL=254
Reply from 10.120.1.2: bytes=32 time=0ms TTL=254

Pinging 10.120.1.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.120.0.2

Pinging 10.120.0.2 with 32 bytes of data:

Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127
Reply from 10.120.0.2: bytes=32 time=0ms TTL=127

Pinging 10.120.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.120.1.2

Pinging 10.120.1.2 with 32 bytes of data:

Reply from 10.120.1.2: bytes=32 time=0ms TTL=254
Reply from 10.120.1.2: bytes=32 time=0ms TTL=254
Reply from 10.120.1.2: bytes=32 time=0ms TTL=254
Reply from 10.120.1.2: bytes=32 time=0ms TTL=254

Pinging 10.120.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.120.1.3

Pinging 10.120.1.3 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.120.1.3: bytes=32 time=0ms TTL=254
Reply from 10.120.1.3: bytes=32 time=0ms TTL=254

Pinging 10.120.1.3:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\> Ping 10.120.0.2

Trying to connect... 10.120.0.2
Connected to 10.120.0.2.
220: welcome to PT FTP server
User name [anonymous]:
230: Username ok, need password
Password:
230: logged in
(anonymous user on)
ftp>
C:\>
```

Рис. 35: Все пингуется, доступ по FTP есть!

Web Browser

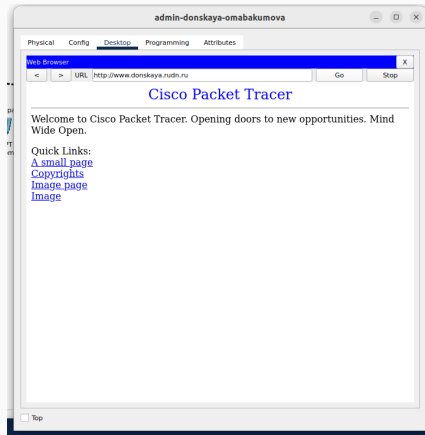


Рис. 36: Страница в Web Browser при запросе www.donskaya.rudn.ru



Рис. 37: Страница в Web Browser при запросе 10.128.0.2

Выводы

Во время выполнения данной лабораторной работы я освоила настройку прав доступа пользователей к ресурсам сети.